Estimating in Building Construction

Chapter 6 Overhead and Contingencies

Overhead

- Home Office Overhead
  - Items that cannot be readily charged to any one project, but represent the cost of operating the construction company
- Job Overhead Cost
  - Overhead expenses resulting in executing a specific project

Home Office Overhead

- Also known as:
  - General overhead
  - Indirect overhead cost

Home Office Overhead

- Rent
- Office utilities
- Office supplies
- Postage
- Insurance on office
- Taxes on office
- Office equipment
- Salaries
  - Executives
  - Accountants
  - Estimators
  - Purchasing dept.
  - Store keepers
  - Secretaries

Home Office Overhead

- Advertizing
- Books & magazines
- Legal fees
- Design costs
- Donations
- Travel
- Vehicles for office staff
- Club and association dues
- Depreciation
  - Office equipment
- See Fig 6.1 Home office overhead and cost allocation

Example 6.1

- Home Office Cost Allocation:

  Anticipated home office costs for fiscal year = $690,580

  Anticipated sales volume for fiscal year = $9,500,000

  Home office cost allocation = 690,580/9,500,00 = 7.3%

Job Overhead

- Also known as:
  - Project overhead
  - Direct overhead
  - Indirect field costs

Job Overhead Includes

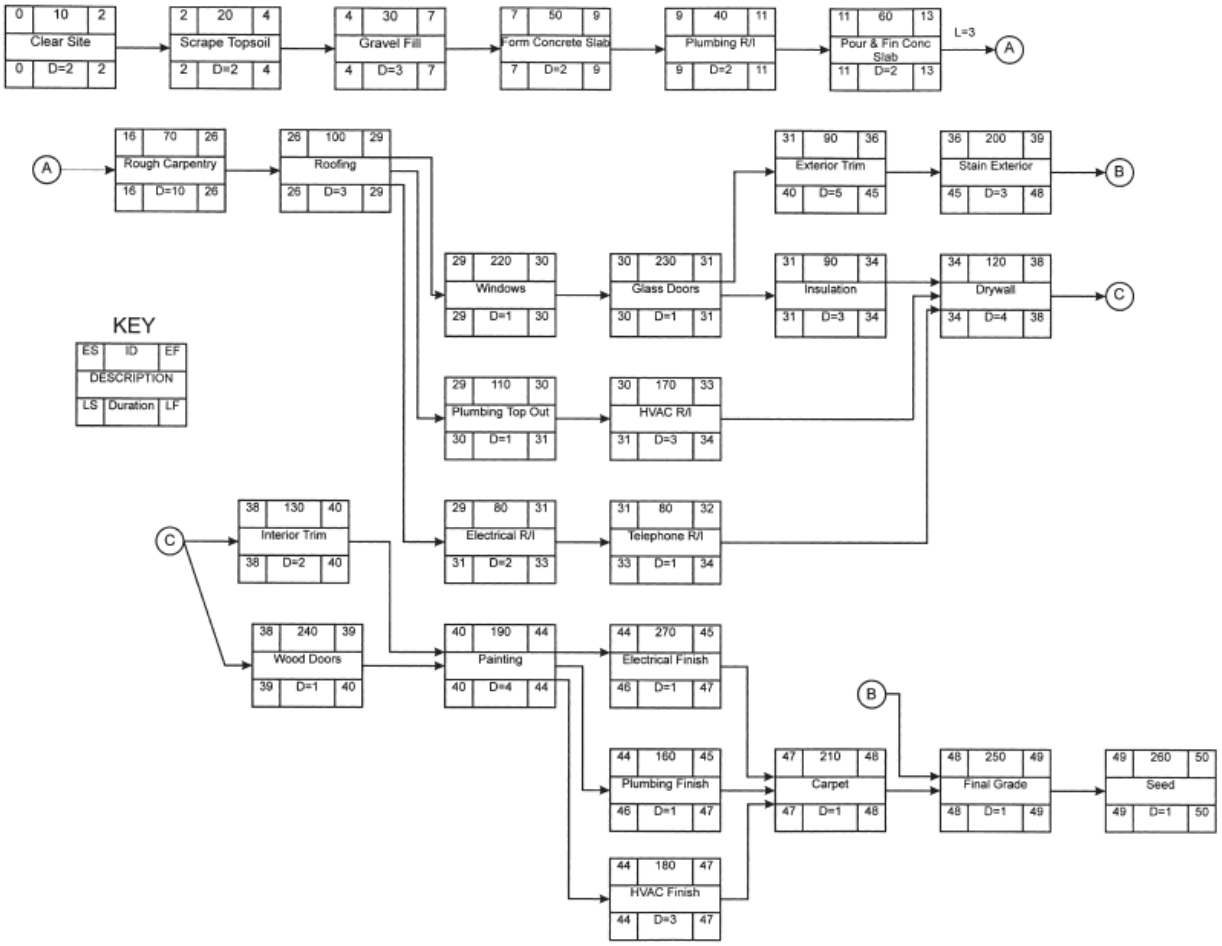
- Salaries
  - Superintendent
  - Assistant superintendent
  - Timekeeper
  - Foremen
  - Security
  - See Fig 6.2 Field staff plan and estimate
- Temporary office
  - Rent
  - Utilities
- Temporary
  - Buildings
  - Barricades (& lights)
  - Enclosures (fence, temporary doors/windows, protection for equipment)

Job Overhead Includes

- Temporary utilities
  - Power
  - Lights
  - Heat
  - Phone
  - Water
- Sanitary facilities
  - Toilets
- Photographs
- Surveys
- Cleanup
- Winter conditions
- Protection of property

Schedule

- Needed to determine job overhead
- See Fig 6.3 Office Building
- Process
  - List all activities
  - Assign durations
  - Sequence the activities
  - Calculate start and finish dates
  - See Fig. 6.4 Activity Lest



Precedence Diagram

Figure 6.5

Contingencies

- For items overlooked or left out
- A sum of money or percentage
- Often used as an excuse for poor estimating practices
- Rational use for contingencies is for price escalation

Checklist

- Estimate of Labor
  - Number of weeks on the project x Weekly rate
  - Persons to be relocated x cost per person
  - Quantity required x productivity rate
- Estimate of Material
  - Quantity required x unit cost
- Estimate of Equipment
  - Duration required x Monthly cost

Read Chapter 1, 9

- Book
  - Aggarwal, A. and Upadhyay, A.K. (2008). Civil Estimating, Costing and Valuation. S.K.Kataria & Sons. New Delhi.

Thank You